

CS 2300 HW 10

- 1) Let us consider 3 sets a,b and c such that a represents all the numbers that are divisible by 3, b represents all the numbers divisible by 5 and c represents all the numbers divisible by 7.

We can solve this question by using the following identity:

$$|a \cup b \cup c| = |a| + |b| + |c| - |a \cap b| - |b \cap c| + |a \cap b \cap c| - |a \cap c|$$

$$|a| = 50000/3 = 16666$$

$$|b| = 50000/5 = 10000$$

$$|c| = 50000/7 = 7142$$

$$|a \cap b| = 50000/\text{lcm}(3,5) = 50000/15 = 3333$$

$$|a \cap c| = 50000/\text{lcm}(3,7) = 50000/21 = 2380$$

$$|b \cap c| = 50000/\text{lcm}(5,7) = 50000/35 = 1428$$

$$|a \cap b \cap c| = 50000/\text{lcm}(3,5,7) = 50000/105 = 476$$

Hence, we can calculate the $|a \cup b \cup c|$ as follows:

$$|a \cup b \cup c| = 16666 + 10000 + 7142 - 3333 - 2380 - 1428 + 476 = 27143$$

Hence, we can say that between 1 and 50000 inclusive, 27143 numbers are divisible by 3 or 5 or 7.

- 2a) Since each shirt has 8 different colors, 6 different sizes, 2 different patterns and 2 different sleeve types, the total number of shirts can be calculated as follows:

$$8*6*2*2 = 192$$

Therefore, we have 192 shirts in total.

- b) Since red and gold color shirts only come in solid (1 pattern) and short sleeves (1 type) and 6 different sizes, and the colors are available in all patterns, types and sizes, then the total number of shirts can be calculated as follows:

$$\text{Total number of red and gold shirts} = 6*2*1*1 = 12 \text{ shirts}$$

$$\text{Total number of other colored shirts} = 6*6*2*2 = 144 \text{ shirts}$$

$$\text{Total number of shirts} = 144 + 12 = 156 \text{ shirts}$$

Thus, we have 156 shirts in total if red and gold colored shirts are only available in solid pattern and short sleeves type.

3a) Since each of the 6 elements of A can map to any of the 8 elements in B, the total number of different functions from $A \rightarrow B$ can be calculated as follows:

$$|B|^{|A|} = 8^6 = 262144$$

Similarly, since each of the 8 elements in B can map to any of the 6 elements in A, so the number of different relations can be calculated as follows:

$$|A|^{|B|} = 6^8 = 1679616$$

b) A relation is a subset of $A \times B$. Thus the number of pairs possible in $A \times B$ can be calculated as follows:

$$|A| \times |B| = 6 \times 8 = 48$$

We can calculate the number of different relations possible from A to B can be calculated by using 2^n , where n represents the number of possible relations.

$$2^{48} = 281474976710656$$

Therefore 281474976710656 relations are possible over here.

c) A function from $A \rightarrow B$ is said to be one-to-one, when each and every element in A is said to map to distinct elements in B, for $|A| \leq |B|$.

So, if we choose 6 elements from B and assign them to 6 distinct elements in A then:

$$P(8,6) = 8! / (8 - 6)! = 8! / 2! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2! / 2! = 20,160$$

Thus, 20,160 functions are one-to-one from A to B.

d) A function from $B \rightarrow A$ is considered to be onto if every element of A is connected by at least one element from B.

Thus, we can calculate the number of onto functions in the following manner:

$$[6!/0!(6-0)!] \cdot 6^8 - [6!/1!(6-1)!] \cdot 5^8 + [6!/2!(6-2)!] \cdot 4^8 - [6!/3!(6-3)!] \cdot 3^8 + [6!/4!(6-4)!] \cdot 2^8 - [6!/5!(6-5)!] \cdot 1^8 + [6!/6!(6-6)!] \cdot 0^8 = 392520$$

4) In order to use the Pigeonhole principle, we know that:

- There are 9 labs
- There is a guarantee that some lab gets 9 chairs

Let us first try to finance the contrapositive of the Pigeonhole principle. Let us say that in order to avoid any lab getting more than 5 chairs, we can only fit in a maximum of 4 chairs in a lab.

So, the maximum number of chairs that can fit in all the 9 labs without violating this condition is:

$$9 * 4 = 36 \text{ chairs}$$

If we add 1 more chair to the total then it becomes 37 chairs in total.

Therefore, as it is impossible to have 37 chairs in 9 labs, given the condition that each lab can fit in only 4 chairs at max, then by Pigeonhole principle, we have proved that there may be some lab which may have at least 5 chairs in it.

5a) Let us first fix mom and dad in the group. Since mom and dad are already chosen so we need to take 5 more people from the remaining 8, so we can calculate the number people to be chosen in the following way:

$$8!/5!(8-5)! = 8!/5!3! = 56 \text{ ways}$$

Thus, there are 56 ways to choose 5 other people.

The number of ways to arrange all 7 people can be calculated in the following manner:

$$7! = 7*6*5*4*3*2*1 = 5040 \text{ arrangements per group}$$

$$\text{Total ways} = 56 * 5040 = 282240 \text{ ways}$$

Therefore, we can choose 7 people from 10 in any given manner provided that mom and dad are included in the group in 282240 ways.

b) Let MD be a block which represents "Mom" and "Dad", so that means, over here we are going to deal with 6 units: MD + 5 other people

Since Mom and dad are already in the group, we can calculate the other 5 people from the remaining 8 in the following ways:

$$8!/5!(8-5)! = 56 \text{ ways}$$

Therefore, the other 5 people can be chosen in 56 ways.

We can calculate the number of arrangements in the following way:

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

In the Mom-Dad block, mom and dad can also be arranged in 2 different ways and hence we can calculate the total number ways in which people can be chosen, provided mom and dad are right next to each other, can be calculated as follows:

$$56 \cdot 720 \cdot 2 = 80640 \text{ ways}$$

Therefore, there are 80640 ways in which 7 people can be chosen from 10, provided that mom and dad are chosen and they remain right next to each other every time.

6) Total number of outcomes when we flip a coin 8 times can be calculated as follows:

$$2^8 = 256 \text{ outcomes}$$

Favorable outcomes:

If we have exactly 4,5,6,7 or 8 heads:

$$\begin{aligned} \text{Total number of favorable outcomes} &= \frac{8!}{4!(8-4)!} + \frac{8!}{5!(8-5)!} + \frac{8!}{6!(8-6)!} + \frac{8!}{7!(8-7)!} + \\ &\frac{8!}{8!(8-8)!} = 70 + 56 + 28 + 8 + 1 = 163 \end{aligned}$$

Therefore, 163 outcomes contain at least 4 heads.

7) Total number of ways in which people are chosen with no restriction = $\frac{22!}{6!16!} = 74613$ people

a) Let us assume that Ann and Beth are 2 of the 12 women that are chosen.

If both Ann and Beth are not included in the unrestricted 6 person committee then we can choose 6 people from 22 in the following way:

$$\frac{20!}{4!(20-4)!} = \frac{20!}{4!16!} = 4845 \text{ ways}$$

The number of ways in which 6 people can be chosen from 22 given that Ann and Beth do not serve together can be calculated as follows:

$$74613 - 4845 = 69768 \text{ ways}$$

Therefore, 6 people can be chosen from 22 given that Ann and Beth do not serve together in 69768 ways.

b) Total number of ways in which all men who can be chosen in the unrestricted 6 person committee = $10!/6!(10-6)! = 10!/6!4! = 210$ ways

Number of ways in which at least one women can be chosen = $74613 - 210 = 74403$ ways

c) Total number of ways in which all women can be chosen in the group = $12!/6!(12-6)! = 12!/6!6! = 924$ ways

Total number of ways in which all men can be chosen in the group = $10!/6!(10-6)! = 10!/6!4! = 210$ ways

Total number of ways in which at least one man and one woman can be chosen = $74613 - 924 - 210 = 73479$.

Therefore, at least one man and one woman can be chosen in 73479 ways.

8a) Full house = 3 cards of 1 rank + 2 cards of another rank.

Let us first choose a rank for 3 of a kind. We can do that in the following way:

$$13!/1!(13-1)! = 13!/1!(12!) = 13$$

Secondly, let us choose 3 suits out of 4 for that rank:

$$4!/1!(3!) = 4$$

Thirdly, choose a different rank for the pair:

$$12!/1!11! = 12$$

Lastly, choose 2 suits out of 4 for that pair:

$$4!/2!(4-2)! = 4!/2!2! = 6$$

Total number of poker hands that are full house = $13 * 4 * 12 * 6 = 3744$ hands

b) Three of a kind: 3 cards of 1 rank + 2 other cards of different ranks.

Let us first choose a rank for 3 of a kind. We can do that in the following way:

$$13!/1!(13-1)! = 13!/1!(12!) = 13$$

Secondly, let us choose 3 suits out of 4 for that rank:

$$4!/3!(1!) = 4$$

Thirdly, choose 2 different cards:

$$12!/2!10! = 66$$

Lastly, choose 1 suit out of 4 for each of the 2 different cards:

$$4*4 = 16$$

Total number of poker hands that are 3 of a kind = $13 * 4 * 66 * 16 = 54912$ poker hands

Therefore, we need 54912 poker hands in order to make a 3-of-a-kind.

c) One pair = One pair of the same rank + 3 cards of different ranks.

Let us first choose a rank for the pair. We can do that in the following way:

$$13!/1!(13-1)! = 13!/1!(12!) = 13$$

Secondly, let us choose 2 suits out of 4 for that rank:

$$4!/2!(2!) = 6$$

Thirdly, choose 3 other ranks from the remaining cards:

$$12!/3!9! = 220$$

Lastly, choose 1 suit out of 4 for each of the 3 cards:

$$4^3 = 64$$

Total number of poker hands that are full house = $13 * 6 * 220 * 64 = 1098240$ hands

Therefore, we need 1098240 poker hands to make one pair.

d) Five different values: It means that we should have 5 cards with different ranks.

Let us first choose 5 distinct ranks from 13 cards. We can do it in the following way:

$$13!/5!(13-5)! = 13!/5!8! = 1287$$

Secondly, choose 1 suit for each rank. We can do it in the following way:

$$4^5 = 1024$$

So, we can the number of poker hands needed for 5 different ranks:

$$1287 * 1024 = 1317888$$

Therefore, we need 1317888 poker hands in order to have 5 different ranks.

9) By looking at the L.H.S, we can say that it represents the number of ways to take out 2 people from a group which consists of $m + n$ people.

Let us assume that a group consists of 2 sub-groups, A and B such that A has m people in the group and B has n people in the group.

Now, let us try and count the number of ways for choosing 2 people by splitting into mutually exclusive cases:

Case 1: Both people are from group A, so the number of ways in which 2 people can be chosen from group A itself can be calculated as follows:

$$\frac{m!}{2!(m-2)!} \quad (1)$$

Case 2: Both people are from group B, so the number of ways in which 2 people can be chosen from group B itself can be calculated as follows:

$$\frac{n!}{2!(n-2)!} \quad (2)$$

Case 3: 1 person is selected from group A and another person is selected from group B. Since we have m choices to choose from group A and n choices to choose from group B, the number ways in which we can select a person from each group can be calculated by multiplying m and n , that is, mn (3).

So, if we combine (1), (2) and (3), we get:

$$\frac{m!}{2!(m-2)!} + \frac{n!}{2!(n-2)!} + mn$$

Therefore, we can say that the LHS and the RHS are trying to find out the same thing, that is, the number of ways in which 2 people can be chosen from a group of $(m+n)$ people, it's just that on the RHS it splits the problem into mutually exclusive cases and then combines everything.

Hence proved, LHS = RHS.

10) **Prove:** $n!/2!(n-2)! * (n-2)!/(k-2)![n-2-(k-2)]! = n!/k!(n-k)! * k!/2!(k-2)!$, where $n \geq k \geq 2$

Let us first take a look at the LHS.

Over here we are basically choosing 2 people from a group of n people: $n!/2!(n-2)!$

Later on, we are choosing k-2 people from the remaining n-2 people to form a group of k people by doing this: $(n-2)!/(k-2)![n-2-(k-2)]!$

Therefore, we can say that, over here we are trying to calculate the number of ways to choose 2 people first from a group of n people and later on we are choosing k-2 more people from the remaining people in order to make a group of k people that includes the first 2 people as well.

Now, let us take a look at the R.H.S.

Over here, we are basically trying to find the number of ways to select k people from a group of n people ($n!/k!(n-k)!$) and later on we are trying to find the number of ways from which we can select 2 people from the original group of k people ($k!/2!(k-2)!$).

Therefore, by looking at the meanings of the LHS and the RHS we can say that basically, both of them are trying to find the same thing, that is, first choosing the group of k people from n people and then selecting 2 people from the group of k people, it's just that they are using different methods to do it. In LHS, we choose the 2 people first and then we find the rest of the group, whereas, in RHS we choose the group first and then we find the 2 people.

Hence proved, LHS = RHS.